

Subject code and title: BDA601 Big Data and Analytics

Subject length: 6 weeks and 12 weeks

Subject Description:

This subject exposes students to techniques and best practices in collecting, storing, extracting, cleaning, manipulating, visualising and analysing structured, semi-structured or unstructured data sets to uncover information which is critical for organisational decisions. Students will be introduced to applications of big data analytics across industry sectors such as segmentation and prediction, recommended systems and targeted marketing, sentiment analysis and operational analytics.

Subject Learning Outcomes (SLO):

SLO a)	Explain and evaluate the V's of Big Data (Volume, Velocity, Variety, Veracity, Valence, and Value)
SLO b)	Identify best practices in data collection and storage, including data security and privacy principles
SLO c)	Apply data science principles to cleaning, manipulation and visualisation of data
SLO d)	Design analytical models based on given problems
SLO e)	Effectively report and communicate findings to an appropriate audience

Delivery Schedule

	Module	Learning Activities	Assessment Progression & Due Date	SLOs addressed
1	Introduction to Big Data and Analytics	- Hands-on activity - Discussion forum - Knowledge check quiz	Familiarise with Assessment 1 brief	a)
2	Data Sourcing and Ingestion	- Hands-on activity - Discussion forum - Knowledge check quiz		b)
3	Big Data Integration and Storage	- Discussion forum - Knowledge check quiz		b)
4	Analytics Engine for Big Data – Apache Spark	- Hands-on activity - Discussion forum - Knowledge check quiz	Assessment 1 Due: Report (30%)	c), d), e)
5	Data Exploration and Cleaning	- Hands-on activity - Discussion forums	Familiarise with Assessment 2 brief	c)
6	Classification	- Hands-on activity - Discussion forums		d)
6	Model Evaluation	- Hands-on activity - Discussion forums		d)
8	Predictive Modelling	- Hands-on activity - Discussion forum	Assessment 2 Due: Source Code and Report (30%)	d)
9	Clustering	- Hands-on activity - Discussion forum	Familiarise with Assessment 3 brief	c), d)
10	Association rules	- Hands-on activity - Discussion forum		d), e)

11	Graph analytics	- Hands-on activity - Discussion forum		d), e)
12	Big data privacy and security	- Hands-on activity - Discussion forum	Assessment 3 Due: Source Code and Presentation (40%)	e)

Time Management:

Below outline the minimum time management expectations for students.

12 Week Delivery:

- 10 Hours per module (one week): Facilitated study: 3 hours / week. Personal Study: 7 hours / week.
- 3 hours facilitated study consists of attending class, responding to facilitator feedback.
- Students are to allocate 7 hours of personal learning. This includes essential time spent on pre-reading and viewing materials, assessment progression and learning activities.

6 Week Delivery:

- 10 Hours per module: Facilitated study: 3 hours / module. Personal Study: 7 hours / module. (Typically 2 modules per week).
- 6 hours facilitated study consists of attending class, responding to facilitator feedback.
- Students are to allocate 14 hours of personal learning. This includes essential time spent on pre-reading and viewing materials, assessment progression and learning activities.

Learning Activities:

There are learning activities in each module with the types of activities listed in the delivery schedule above. These activities are an essential component of the subject and they are designed to prepare you for your graded assessment tasks. These learning activities will help you to build your understanding and skills whilst also collaborating with your peers.

***Detailed information of each activity can be found in their corresponding Module page in Blackboard.**